Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

ANSWER KEY

Section 1: Reciprocal Trigonometric Functions

1 Simplify the trigonometric expression $\frac{\sec x - \cos x}{\tan x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\cot x$

d. $\csc x$

$\sec x - \cos x = \frac{1}{\cos x} - \cos x = \frac{1 - \cos^2 x}{\cos x} = \frac{\sin^2 x}{\cos x}$. Dividing by $\tan x = \frac{\sin x}{\cos x}$ gives $\frac{\sin^2 x}{\cos x} \cdot \frac{\cos x}{\sin x} = \sin x$.

2 Find the exact value of $\csc\left(\frac{5\pi}{6}\right)$.

a. 2 ✓

b. -2

c. $\frac{2}{\sqrt{3}}$

d. $-\frac{2}{\sqrt{3}}$

$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$. Since $\csc\theta = \frac{1}{\sin\theta}$, $\csc\left(\frac{5\pi}{6}\right) = \frac{1}{1/2} = 2$.

3 Which identity relates $\cot\theta$ and $\csc\theta$ for all defined angles?

a. $1 + \cot^2\theta = \csc^2\theta$ ✓

b. $1 + \csc^2\theta = \cot^2\theta$

c. $\cot^2\theta - \csc^2\theta = 1$

d. $\csc\theta = 1 + \cot\theta$

Dividing $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ yields the fundamental Pythagorean identity $1 + \cot^2\theta = \csc^2\theta$.

Section 2: Compound Angle Addition Formulas

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4 Use compound angle identities to find the exact value of $\cos(75^\circ) = \cos(45^\circ + 30^\circ)$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d. $\frac{\sqrt{2}-\sqrt{6}}{4}$

$\cos(A+B) = \cos A\cos B - \sin A\sin B = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$.

5 Simplify $\sin(x+y)\cos y - \cos(x+y)\sin y$.

a. $\sin x$ ✓

b. $\cos x$

c. $\sin(x+2y)$

d. $\cos(x-y)$

Using the subtraction formula $\sin(A-B) = \sin A\cos B - \cos A\sin B$ with $A = x+y$ and $B = y$, this simplifies to $\sin((x+y)-y) = \sin x$.

6 If $\tan A = 2$ and $\tan B = 3$, what is $\tan(A+B)$?

a. -1 ✓

b. 1

c. 5

d. $\frac{5}{7}$

$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B} = \frac{2+3}{1-2(3)} = \frac{5}{1-6} = \frac{5}{-5} = -1$.

Section 3: Double Angle Formulas

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7 If $\sin\theta = \frac{3}{5}$ and θ is acute, what is $\cos(2\theta)$?

a. $\frac{7}{25}$ ✓

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{16}{25}$

$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{3}{5}\right)^2 = 1 - 2\left(\frac{9}{25}\right) = 1 - \frac{18}{25} = \frac{7}{25}$.

8 Under the same conditions ($\sin\theta = \frac{3}{5}$ acute), what is $\sin(2\theta)$?

a. $\frac{24}{25}$ ✓

b. $\frac{7}{25}$

c. $\frac{12}{25}$

d. $\frac{6}{5}$

$\cos\theta = \sqrt{1 - 9/25} = \frac{4}{5}$. Then $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

9 Express $\cos^2 x$ in terms of a first-power cosine function.

a. $\frac{1+\cos(2x)}{2}$ ✓

b. $\frac{1-\cos(2x)}{2}$

c. $\frac{1+\sin(2x)}{2}$

d. $1 - \sin^2 x$

Rearranging $\cos(2x) = 2\cos^2 x - 1$ gives $2\cos^2 x = 1 + \cos(2x) \Rightarrow \cos^2 x = \frac{1+\cos(2x)}{2}$.

Section 4: Linear Trigonometric Combinations (R-form)

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10 Express $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$ with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

a. $5\sin(x + 0.927)$ ✓

b. $5\sin(x + 0.643)$

c. $7\sin(x + 0.927)$

d. $25\sin(x + 0.927)$

$R = \sqrt{3^2 + 4^2} = 5$. Expand $R\sin(x+\alpha) = R\cos\alpha\sin x + R\sin\alpha\cos x$. Thus $R\cos\alpha = 3$ and $R\sin\alpha = 4 \Rightarrow \tan\alpha = 4/3 \Rightarrow \alpha = \arctan(4/3) \approx 0.927$ rad.

11 What is the maximum value of the function $f(x) = 5\cos x - 12\sin x + 3$?

a. 16 ✓

b. 13

c. 10

d. 20

The combination $5\cos x - 12\sin x$ has amplitude $R = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = 13$. The maximum value is $13 + 3 = 16$.

12 Solve $\sin(2x) = \cos x$ for $x \in [0, 2\pi)$.

a. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

b. $x = \frac{\pi}{6}, \frac{5\pi}{6}$

c. $x = 0, \pi$

d. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$. $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$.

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